Project 2:

Data Analysis Plan

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Table of Contents

Data Management and Cleaning 2
General Analytical Framework 2
Details of the Analyses 2
Power Analysis 2

Data Management and Cleaning

Data will be collected from a study conducted at the Rocky Mountain Alzheimer's Disease Center (RMADC) at the University of Colorado Anschutz Medical Campus and will include demographic and clinical variables, plasma inflammatory markers, cognitive assessments, structural neuroimaging, and amyloid PET measures collected at baseline and one-year follow-up.

Continuous outcomes will be evaluated for outliers, missingness, and distributional assumptions using graphical summaries and descriptive statistics, with transformations applied if needed. Participants excluded due to missing exposures or outcomes will be compared with the full cohort to evaluate potential bias. Sensitivity analyses will assess whether results differ between the full dataset and the analytic subset.

General Analytical Framework

Linear regression models will be used to estimate associations between inflammatory markers and one-year changes in cognitive performance and AD-signature cortical thickness, allowing adjustment for covariates and testing of interaction effects.

Because preliminary analyses indicated weak correlation between the cognitive and cortical thickness outcomes, models will be estimated separately for each outcome.

Statistical significance will be assessed using two-sided tests with a Bonferroni style-adjusted significance level of $\alpha = 0.005$ within each hypothesis family.

Because inflammatory markers and outcomes are expected to be correlated, results will be summarized by evaluating consistency in effect size and direction across related markers and outcomes rather than relying on isolated statistically significant findings.

Details of the Analyses

Primary outcomes are one-year changes in cognitive performance and AD-signature cortical thickness (five outcomes total), with higher values indicating worsening cognition or cortical thinning.

Primary exposures include circulating inflammatory markers measured in plasma at baseline and at one-year follow-up. All models will adjust for age, sex, APOE genotype, baseline Clinical Dementia Rating (CDR), and baseline outcome value. Additional health-related variables associated with inflammatory processes (e.g., BMI, cardiovascular risk factors, NSAID use, immune-related health conditions) will be evaluated as potential confounders in sensitivity analyses.

For Hypothesis H1a, linear regression models will evaluate associations between baseline inflammatory markers and one-year outcome changes. For Hypothesis H1b, models will evaluate whether one-year changes in inflammatory markers are associated with outcome changes.

For Aim 2, models will extend Aim 1 by including amyloid deposition and an inflammation x amyloid interaction term to assess whether amyloid modifies the association between inflammation and the outcomes.

Amyloid deposition will be dichotomized using an SUVR threshold of 1.10, which demonstrated 97% sensitivity and 100% specificity for amyloid positivity in previous analyses. An indicator variable will therefore be coded as binary (low vs high amyloid).

Summary of Analyses

Aim	Hypothesis	Primary Outcome	Primary Exposure	Covariates
1	H1a	1-yr Δ in CVLT, Benson Figure, MST, and Store Recall (cognitive measures); 1-yr Δ in AD-signature cortical thickness (brain structure)	Baseline inflammatory markers (cytokines: IL-6, TNF-α, chemokines:MCP-1, Eotaxin-1)	Age, sex, APOE genotype, baseline CDR, baseline outcome
1	H1b	Same outcomes	Change in inflammatory markers (baseline → year-1)	Age, sex, APOE genotype, baseline CDR, baseline outcome
2	H2	Same outcomes	Inflammation × baseline amyloid interaction	Age, sex, APOE genotype, baseline CDR, baseline outcome

Power Analysis

Power calculations were conducted assuming a final analytic sample size of N = 175.

Because inflammatory markers are classified into two biologically related groupings (cytokines and chemokines), and preliminary data indicated strong correlation between representative markers (IL-6 and MCP-1; $\rho = 0.93$, 95% CI: 0.86–0.97), analyses were treated as evaluating two correlated groups rather than four independent predictors.

Accordingly, multiple testing within each hypothesis was defined across five outcomes and the two inflammatory groups (5 outcomes x 2 markers = 10 tests), and power calculations used a Bonferroni style-adjusted two-sided $\alpha = 0.005$ (0.05/10), with $\alpha = 0.0025$ (0.05/20) evaluated as a conservative sensitivity analysis on the full Bonferroni-adjusted alpha.

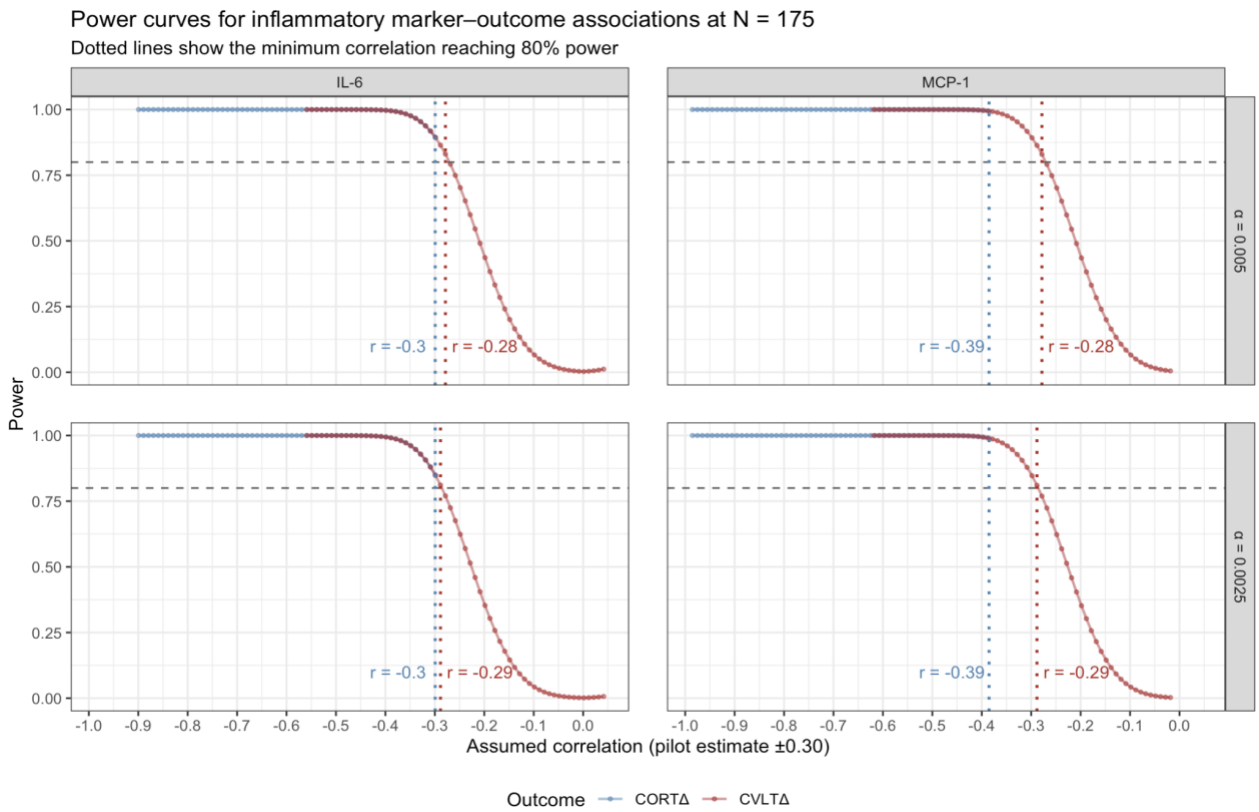

Figure 1. Power to detect associations between inflammatory markers and outcomes at N = 175. *Power curves illustrate the estimated power to detect associations between inflammatory markers (IL-6 and MCP-1) and one-year changes in cognitive performance (CVLTΔ) and AD-signature cortical thickness (CORTΔ), evaluated across a range of plausible correlations informed by preliminary data (±0.30 around pilot estimates). The dashed horizontal line represents 80% power, and vertical dotted lines indicate the minimum detectable correlation for each outcome and α level.*

For Aim 1, although final analyses will use covariate-adjusted linear regression, power was approximated using correlation-based methods (powertools::corr.1samp in R). Effect sizes were based on preliminary correlations between IL-6, MCP-1, and change outcomes (CVLT and AD-signature cortical thickness), and power was evaluated across ranges defined as ±0.30 around each correlation estimate.

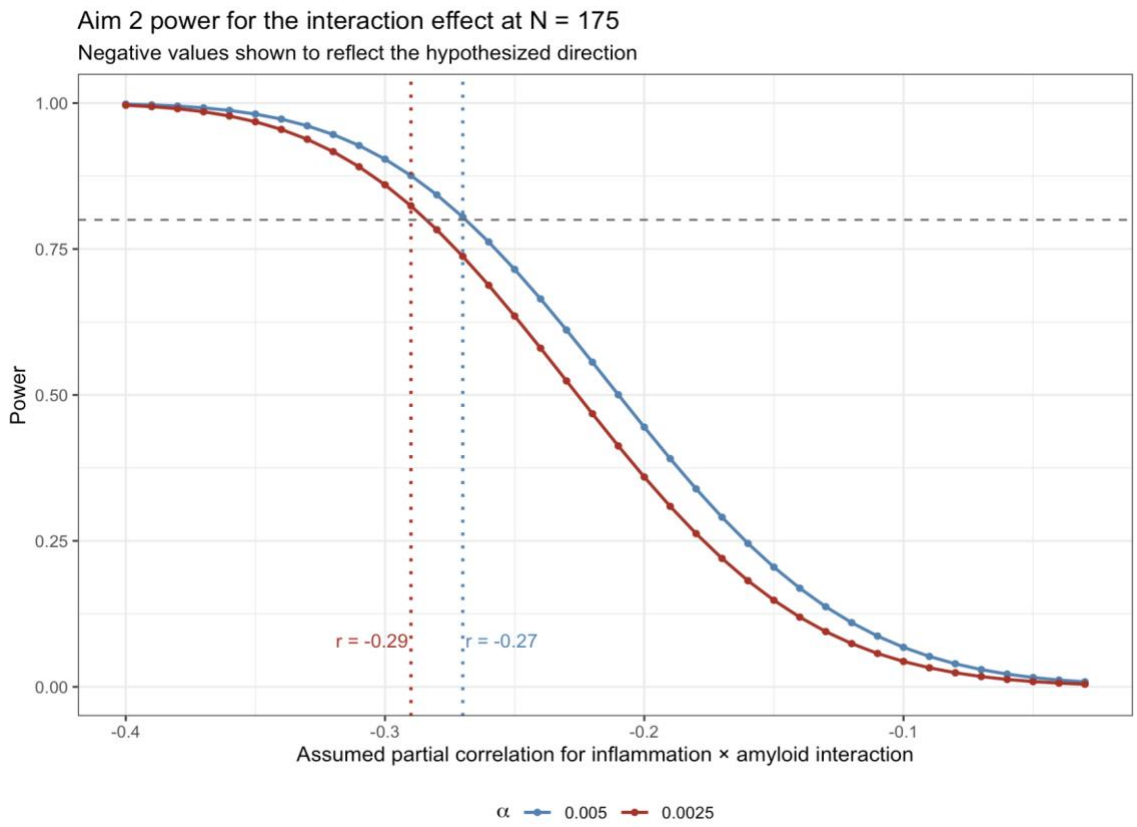

Figure 2. Power to detect the inflammation × amyloid interaction at N = 175. *Power curves display the estimated power to detect the interaction between peripheral inflammation and amyloid deposition as a function of the assumed partial correlation for the interaction term. Negative values are shown to reflect the hypothesized direction of association (greater inflammation and amyloid associated with worse outcomes). The dashed horizontal line represents 80% power, and vertical dotted lines indicate the minimum detectable interaction effect size at each α level.*

For Aim 2, because pilot data did not include amyloid deposition, power was evaluated across a plausible range of partial correlations using powertools::mlrF.partial in R. We assumed the reduced model included six predictors (covariates plus inflammation and amyloid; p = 6), and the full model included the interaction term (q = 1).

Power results (Figures 1–2) indicate that correlations of approximately |r| ≈ 0.28–0.39 are required to achieve 80% power for Aim 1, and partial correlations of approximately 0.27–0.29 for Aim 2, depending on alpha. These results suggest the proposed sample size is sufficient to detect moderate effects but may be underpowered for small effects.